

# Evaluation of $I = \int_0^1 \cos(x^2 + e^x) dx$

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## 1 Introduction

We are looking to evaluate

$$I = \int_0^1 \cos(x^2 + e^x) dx.$$

No elementary antiderivative exists, but we can obtain an exact expression in the form of a double series.

## 2 Power series expansion

For any real number  $u$ ,

$$\cos u = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} u^{2n}.$$

By setting  $u = x^2 + e^x$ , we have

$$\cos(x^2 + e^x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (x^2 + e^x)^{2n}.$$

## 3 Binomial expansion

For each  $n$ , let us expand  $(x^2 + e^x)^{2n}$  using the binomial theorem:

$$(x^2 + e^x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} (x^2)^k (e^x)^{2n-k} = \sum_{k=0}^{2n} \binom{2n}{k} x^{2k} e^{(2n-k)x}.$$

## 4 Series/integral interchange

The series converges normally on  $[0, 1]$ , so we can integrate term by term:

$$I = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \sum_{k=0}^{2n} \binom{2n}{k} \underbrace{\int_0^1 x^{2k} e^{(2n-k)x} dx}_{I_{k,n}}.$$

## 5 Exact calculation of $I_{k,n}$

Let  $m = 2k$  and  $a = 2n - k$ . We must evaluate

$$I(m, a) = \int_0^1 x^m e^{ax} dx.$$

- If  $a = 0$  (i.e.,  $k = 2n$ ):  $I(m, 0) = \frac{1}{m+1} = \frac{1}{4n+1}$ .
- If  $a \neq 0$ , we integrate by parts  $m$  times or use the explicit formula

$$\begin{aligned} I(m, a) &= \frac{m!}{(-a)^{m+1}} \left( 1 - e^a \sum_{j=0}^m \frac{(-a)^j}{j!} \right) \\ &= \frac{(2k)!}{(k-2n)^{2k+1}} \left( 1 - e^{2n-k} \sum_{j=0}^{2k} \frac{(k-2n)^j}{j!} \right). \end{aligned}$$

## 6 Exact expression of the integral

By substituting both cases into the double sum, we obtain the exact representation:

$$\begin{aligned} I &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left[ \binom{2n}{2n} \frac{1}{4n+1} \right. \\ &\quad \left. + \sum_{\substack{k=0 \\ k \neq 2n}}^{2n} \binom{2n}{k} \frac{(2k)!}{(k-2n)^{2k+1}} \left( 1 - e^{2n-k} \sum_{j=0}^{2k} \frac{(k-2n)^j}{j!} \right) \right]. \end{aligned} \quad (1)$$

This expression is exact and only involves factorials, binomial coefficients, and the exponential  $e$ .

## 7 Numerical evaluation

### 7.1 Direct evaluation

Using an adaptive quadrature method (Gauss-Kronrod), we find

$$I \approx -0.303666963947415 \quad (\text{estimated error} \sim 10^{-15}).$$

### 7.2 Evaluation of the exact series

The naive calculation of (1) in double precision fails quickly: the terms  $e^{2n-k}$  and the exponential sums are very large, and their subtraction causes catastrophic cancellation. For  $n \geq 10$ , the numerical results blow up.

Using multi-precision arithmetic (50 decimal places) and a robust numerical integration of each  $I_{k,n}$  (rather than the unstable closed formula), the double series converges to:

$$I = -0.303666963947415171254014036015 \dots$$

The two approaches match up to machine precision, validating the exact formula.

## 8 Retained approximate value

$$\int_0^1 \cos(x^2 + e^x) dx \simeq -0.303666963947415$$

with an uncertainty of less than  $10^{-14}$ .

### Remark

The expression (1) provides an **exact value** of the integral, although it cannot be reduced to an elementary constant. For practical calculations, direct numerical integration is preferred.